

KAUSHIK MASLEKAR

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<https://github.com/KaushikMaslekar> portfolio: kaushikmaslekar.vercel.app

SUMMARY

Final-year B.E. student in Artificial Intelligence & Data Science specializing in backend engineering and distributed systems. Experienced in building REST APIs, event-driven services, and AI-powered applications using Java, Spring Boot, Python, Kafka, Redis, Docker, and AWS. Strong foundation in RAG pipelines, semantic retrieval, computer networks, and scalable backend architecture.

TECHNICAL SKILLS

Languages: Java, Python, C++
Backend & Distributed Systems: Spring Boot, REST APIs, Microservices, Kafka, Redis, Event-Driven Architecture, Multithreading, Load Balancing
AI/ML: Machine Learning, LLM, Neural Networks, RAG, LangChain, Pandas, NumPy
Databases: MySQL, PostgreSQL, MongoDB
Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD
Testing & Observability: JUnit, Mockito, Postman, Prometheus, Grafana

EXPERIENCE

AI/ML Intern — YBI Foundation

Feb 2026 – May 2026

- Developed a Retrieval-Augmented Generation (RAG) application using Python and LangChain for document-grounded question answering over large document collections.
- Implemented document chunking, embedding generation, and semantic retrieval to improve retrieval relevance and answer quality.

KEY PROJECTS

Full-Stack RAG Chat Platform | Next.js, Spring Boot, FastAPI, ChromaDB

- Architected a full-stack RAG application with a Next.js frontend, Spring Boot API gateway, and FastAPI retrieval service communicating through SSE streaming.
- Implemented JWT authentication and API rate limiting at the gateway and built a ChromaDB-based retrieval pipeline with page-aware chunking and cross-encoder reranking.
- Evaluated retrieval and generation quality on a golden evaluation set, achieving 83.3% Recall@5, 0.83 MRR, and 100% correct abstention, with 76 ms average end-to-end query latency and 4.2 citations per answer.

Encrypted Traffic Classifier | Java, Spring Boot, Machine Learning, Packet Analysis

- Designed an encrypted traffic classification platform for network visibility and threat detection without decrypting TLS traffic.
- Built a flow-analysis pipeline covering packet capture, flow reconstruction, metadata extraction, and JA3 fingerprinting.
- Trained an ML-based traffic classifier and integrated it with a scalable Spring Boot backend for traffic classification.

MachinoCare — AI-Powered Predictive Maintenance Platform | FastAPI, ESP32, Streamlit, PostgreSQL

- Built an end-to-end predictive maintenance platform combining IoT sensor data, ML-based anomaly detection, and real-time monitoring.
- Developed a FastAPI backend to ingest ESP32 vibration data through REST/WebSockets, backed by PostgreSQL and a Streamlit dashboard for live diagnostics.

ACHIEVEMENTS

- 1st Place — Mindstrom Hackathon 2026: Developed MachinoCare, an IoT-based machine health monitoring and predictive maintenance platform.
- 2nd Place — Mindstrom Hackathon 2025: Developed an IoT-based fire detection system.
- ₹50,000 Project Funding: Secured funding for an IoT-based plant monitoring system.

EDUCATION

B.E. in Artificial Intelligence and Data Science (Pursuing)
Marathwada Mitra Mandal's College of Engineering, Pune | Aug 2024 – Present
Expected Graduation: 2027 | CGPA: 8.6/10